
Effective Rank as a Diagnostic for Representation Bottlenecks and Compression in Language Models

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Abstract

Large Language Models (LLMs) are commonly trained in overparameterized ways and then compressed afterwards, often via post-training quantization. The relevant diagnostic baselines include LLaMA.cpp quantization [1] and Activation-aware weight quantization(AWQ) [2]. But it remains unclear how much of the parameter space has been occupied by the training model in its intermediate representations and weights. Dohare et al. (2024) [3] showed that effective rank for entropy-of-spectrum has been a representational diagnostic tool in modern deep learning. And effective rank has been a powerful tool for diagnosing and guiding quantization in practice, proposed by Roy and Vetterli (2007) [4]. The our main question for addressing is: Can effective rank of each layer for representations and weights reliably diagnose representation bottlenecks and predict compressional potential and lower-bit quantization performance? Our method mainly focus on evaluate the effective rank of the representation spectrum and weight spectrum on language models Qwen 3 0.6B, and then use the effective rank to guide diagnosis and compression, followed by comparing results to uncompressed model and visualizing the effective rank across layers to figure out patterns. The key takeaways are the empirical guidelines of the power of effective rank as a diagnostic for representation bottlenecks and compression in language models, and actionable compression rules. Our further impact is to provide mathematical proof of effective rank implying good compression performance under certain assumptions to justify our empirical findings.

1 Introduction

Background:

Recent progress in large language models consolidates a common design pattern: models are built overparameterized and compressed after training. This trend appears both in practice, where we widely adopt post-training quantizations, and in theory, where overparameterization is often associated with smoother optimization and training dynamics. This raises a basic but important question: how much of this parameter space is actually utilized by a trained model, in both intermediate representations and weights?

If trained networks use fewer effective directions than their architecture framework suggests, the spectra and effective rank of representations and weights may provide a simple measure, and patterns across layers may reflect how computation and information flow inside the model. Studying this

effective dimensionality may therefore help connect interpretation and compression: it can suggest which layers are capacity-limited, which are overparameterized, and where low-rank approximation or lower-bit quantization is possible.

Inspirations:

- Importance matrix quantization[1].
- AWQ (Activation-Aware Weight Quantization)[2].
- Spectral entropy and effective rank[4].

2 Research Plan

Consider one layer of a neural network:

$$y = \phi(Wu), \quad u = \text{Norm}(x),$$

where x is the activation from the previous layer, u is the normalized activation, Norm is the normalization used, W is the weight matrix, and ϕ is the nonlinearity.

We propose the following experiment plan:

1. Choose a small language model, such as Qwen 3 0.6B.
2. Collect a sample of normalized activations $[u_1, u_2, \dots]$ from data and stack them into a matrix U . Also, collect the trained weight matrix W .
3. Compute the spectra of the activation covariance and the weight matrix:

$$U^T U = Q \Lambda Q^T, \quad W = U \Sigma V^T,$$

which gives the eigenvalues $[\lambda_1, \lambda_2, \dots]$ and singular values $[\sigma_1, \sigma_2, \dots]$.

4. Compute spectral entropy and effective rank as a measure of complexity and dimensionality:

$$p_i = \frac{\lambda_i}{\sum_j \lambda_j} \quad \text{or} \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j},$$

$$H(p) = - \sum_i p_i \log p_i, \quad r_u \text{ or } r_w = e^{H(p)}.$$

where r_u is the effective rank of the representation spectrum, and r_w is the effective rank of the weight spectrum.

5. Use r_u and r_w to guide diagnosis and compression:
 - (a) r_u represents the intrinsic dimension of the input representation, while r_w represents the intrinsic capacity of the operator.
 - (b) A large r_u with a small r_w may indicate a bottleneck. A small r_u with a small r_w may indicate a well-matched layer that could be compressed. A small r_u with a large r_w may suggest that the weights can be compressed.
 - (c) Test direct low-rank factorization of Wu , for example:

$$Wu \approx W V_{r_u} V_{r_u}^T u.$$

- (d) Test allocating quantization bits based on effective rank, for example, using lower-bit quantization (such as Q3) for layers with low effective rank and higher-bit quantization (such as Q5) for layers with high effective rank.
6. Evaluate the compressed model and diagnosis:
 - (a) Compare speed and compression results against the uncompressed model, including model size, tokens per second, and perplexity.
 - (b) Visualize how r_u and r_w change across layers and see if there are patterns.

3 Author Contribution Plan

- Peng: code the above experiment; Brainstorming for better diagnosis and compression methods; discuss proof.
- Fergie: Help with code and experiments; writing; mathematical proof.

AI usage statement

ChatGPT is used for polishing writing and helping brainstorm ideas.

References

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